

Mohamed Amine Hajji

Software Engineer

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Summary

Recently graduated Software Engineer with hands-on experience in full-stack development, machine learning, and mathematical optimization. Strong interest in algorithms and problem-solving, with experience building production applications and data-driven solutions. Quick to learn new technologies and adapt to new technical domains.

Education

Higher Institute of Computer Science (ISI), National Engineering Degree in Software Engineering – Ariana, Tunisia Sept 2023 – June 2026

Higher Institute of Computer Science (ISI), Bachelor's Degree in Computer Science – Ariana, Tunisia Sept 2020 – June 2023

Borj Louzir High School, Baccalaureate (Honors) in Mathematics – Tunisia Sept 2016 – June 2020

Experience

Data & AI Engineer Intern - End-of-Studies Project (PFE), Drivago – Tunisia Feb 2026 – June 2026

- Designed a machine learning pipeline to predict candidate driving-lesson hours from historical candidate data.
- Designed and developed a 7-step digital candidate assessment application with Angular and a Three.js 3D cockpit simulator, feeding candidate readiness scores into the hours-prediction model to calibrate its estimates.
- Built an AI scheduling engine (Python/FastAPI, Google OR-Tools CP-SAT) as a standalone service called by the platform's backend, solving full schedules for 50 students, 12 instructors, and a 3-week horizon in about 1.2 seconds with 100% session coverage.
- Trained a churn-prediction model reaching 0.875 ROC-AUC on the platform's main subscription tier to flag at-risk customers before subscription lapse.
- Built Looker Studio dashboards on PostgreSQL data sources covering 600+ partner driving schools, giving the product team visibility into revenue, feature adoption, and churn signals.
- Profiled and optimized the Angular scheduling calendar's change-detection path using Chrome DevTools CPU profiling, cutting average interaction latency by 75% and worst-case latency by 87%.

Full-Stack Software Engineer - Part-Time, DeepVolt – Ariana, Tunisia July 2023 – present

- Migrated the platform's heatmap rendering from GeoJSON to a vector-tile pipeline (using Tippecanoe, hosted on S3), cutting load times by 75-85% (low-zoom: 8-12s to 1-2s).
- Led the adoption of Martin, an open-source Rust-based PostGIS tile server: wrote the PostGIS queries and deployed it to add real-time filtering and clustering across heatmap layers.
- Built configurable competitor, POI, and traffic filters (Angular frontend, Node.js/PostgreSQL backend) that let users fine-tune how the site-scoring algorithm weighs nearby competition, points of interest, and road traffic.
- Optimized data fetching from the TomTom API and integrated a purchased dataset of 100,000+ EV charging stations across France into the platform's map layers.
- Supported the data science team with data cleaning and web scraping to build training datasets, using QGIS to visualize and validate geospatial data.

- Worked on the front-end of a side project for DeepVolt's Istanbul expansion, integrating the team's AI-recommended charging-station locations and connecting the UI to their backend endpoints.

Front-End Developer Intern (End-of-Studies Project), DeepVolt – Ariana, Tunisia Feb 2023 – June 2023

- Built the front-end of DLIA (DeepVolt Location Intelligence Assistant), a tool that helps identify the best locations for EV charging stations, including JWT-based authentication and an admin panel for managing users and companies.
- Built the map exploration experience: point-of-interest markers powered by the TomTom API, a location details panel, and toggleable heatmap layers for area potential score, population density, income, and EV registrations.
- Built the multi-location comparison feature, letting users compare several candidate sites side by side, and integrated the consumption-prediction flow into the UI so users could predict a site's expected EV charging demand.

Projects

Gender Classification on the Adience Dataset

Deep learning model for gender classification from facial images, built with a teammate for an advanced AI course.

- Built a face-detection preprocessing pipeline (OpenCV Haar Cascade) and fine-tuned EfficientNet-B0 via transfer learning in PyTorch.
- Reached 96% test accuracy on 3,247 held-out images from a 16K-image training set.

Research Portal Web Application

A full-stack web app for sharing and exploring research papers, with role-based access and an integrated research chatbot.

- Built with Angular, Angular Material, Node.js/Express, and PostgreSQL, with role-based access for students, researchers, and general users.
- Integrated a research chatbot using Hugging Face APIs for contextual Q&A on each paper.

Multitasking Scheduling Algorithm Simulation System

A simulation platform for visualizing classic multitasking scheduling algorithms, built in a team of 5.

- Implemented FIFO, Round Robin, and SRT scheduling algorithms in C.
- Built a graphical interface with Raylib and Raygui for real-time visualization of scheduling behavior.

Skills

Languages: Python, TypeScript, JavaScript, C, C++, Rust

Frameworks & Tools: Angular, Node.js, Express, FastAPI, Three.js, Mapbox, TomTom API, Martin, Tippecanoe, Git

Data & ML: PostgreSQL, PostGIS, MySQL, MongoDB, PyTorch, Scikit-learn, XGBoost, Google OR-Tools (CP-SAT), Looker Studio, QGIS

Cloud & DevOps: AWS (EC2, RDS, S3), GitHub Actions (CI/CD)

Achievements

- Meta Hacker Cup 2022 — Reached Round 2 (Global Rank: 2993)
- CodinGame Top 0.6% Global Ranking

Volunteer

- Freeways ISI — Problem Solving Department

Languages

Arabic, English, French